



Indian Institute of Technology  
Kharagpur

S&P Global

CRISIL

# TREMOR

## Text-driven Risk Engine for Market Observation & Response

Converting unstructured financial news into explainable risk signals, market scenarios, and credit risk insights for faster, data-driven decisions.



Real-time  
Event Understanding



Actionable  
Risk Signals



Market & Credit  
Impact Analysis



Explainable  
& Transparent





# The Problem & Our Approach

Financial markets react to real-world events. The challenge is to convert the massive, unstructured news and social media flow into timely, explainable and actionable risk insights.



## The Problem

Global events, policy decisions and supply chain shocks move markets, but the information is unstructured, noisy and overwhelming.

### A constant stream of unstructured information



**News (Reuters)** Feb 24, 2022  
"Putin announces military assault against Ukraine in surprise speech"



**News (Financial Times)** Feb 24, 2022  
"European stocks at risk if Russia invades Ukraine"



**News (WSJ)** Feb 24, 2022  
"Oil On The Boil: Brent races to \$105 amid Russia-Ukraine hostilities"



**Social Media (Twitter)** Feb 24, 2022  
"Long \$oil here. Supply disruption looks real."

### Key challenges for risk teams



**High volume & noise**  
Tens of thousands of news articles and social media posts daily across sources and languages.



**Slow and manual analysis**  
Analysts spend significant time reading, tagging and interpreting events.



**Hard to quantify impact**  
Linking events to market and credit impact requires domain expertise and structured analysis.



**Lack of transparency**  
Black-box signals are hard to trust and act on in regulated environments.



## Our Approach – TREMOR

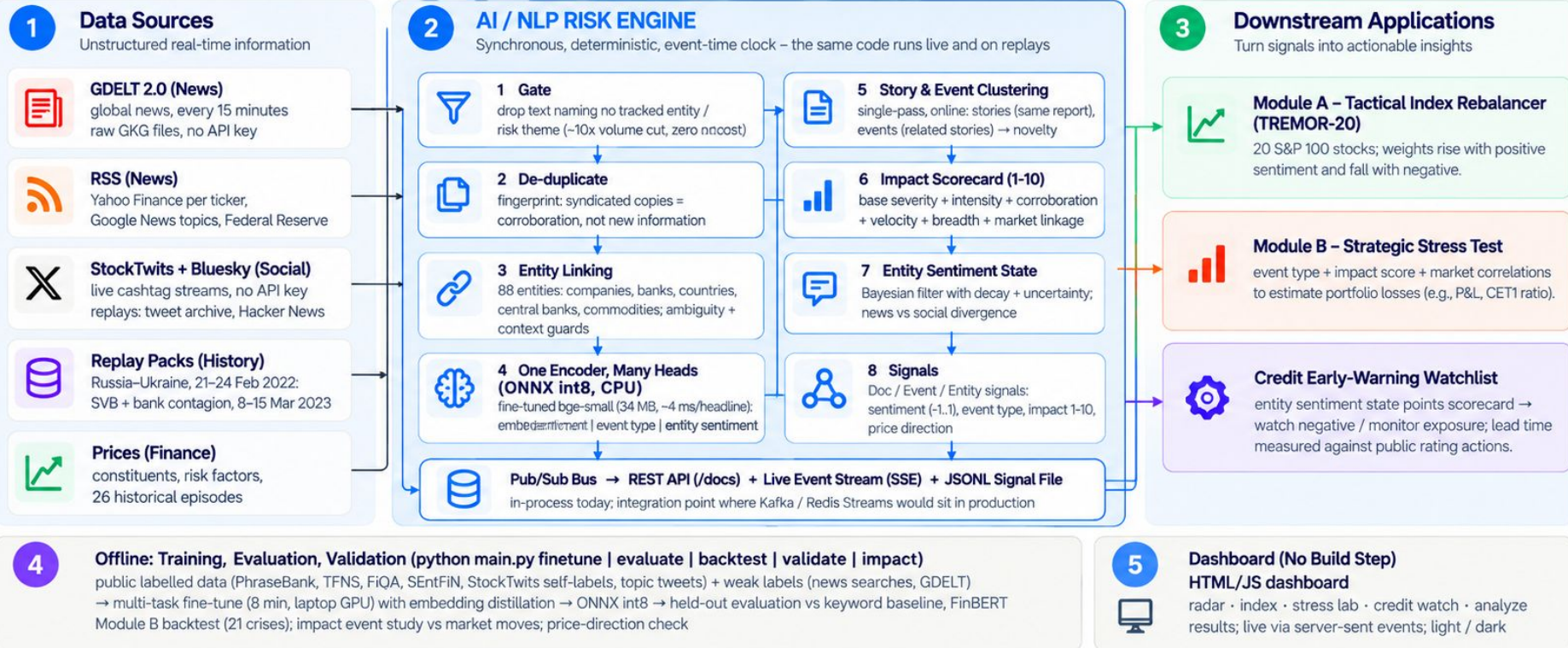
Convert unstructured news and social media into explainable risk signals, market scenarios and credit risk insights.





# TREMOR – System Architecture

From unstructured news and social media to structured risk signals, used by a tactical index, strategic stress test and a credit early-warning watchlist.





# Implementation Highlights – Core Components of TREMOR

A modular pipeline that converts unstructured news and social media into risk signals, market scenarios and credit risk insights.

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## NLP Engine – From News to Structured Events

Detects and classifies financial events with context using a fine-tuned language model.



**News & Social Media**  
(Reuters, FT, WSJ, StockTwits, ...)



**Preprocessing**  
Cleaning, de-duplication,  
segmenting, source filtering



**Event Detection & Classification**  
(what, when, where, type)

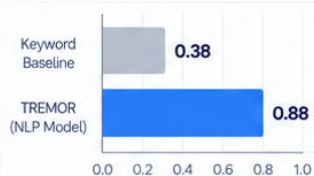


**Entity & Risk Factor Mapping**  
(sector, region, asset class)



**Structured Events**  
(for downstream modules)

### Event Type Classification (Macro F1)



- 39,242 documents processed
- 307 docs/second
- 1.5 – 3 minutes total processing time
- ~4.2 ms per headline (CPU)
- 34 MB model size (ONNX)

2

## TREMOR-20 – Sentiment-weighted Index

Converts structured events into a time-series index capturing market risk factor movements.

### TREMOR-20 vs Naive Baseline – Capture of Market Shocks



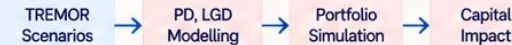
### Captures key market risk factors

- Equity Markets
- Interest Rates
- Commodities
- Credit Spreads
- FX
- Volatility

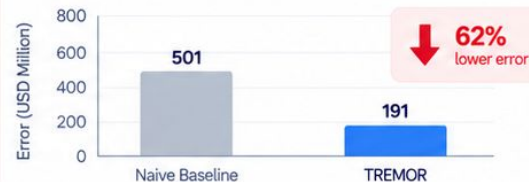
3

## Credit Stress Test – From Signals to Impact

Translates market scenarios into potential credit losses and capital impact for financial institutions.



### Mean P&L Error vs Naive Baseline



### Portfolio & Capital Impact (Ukraine Escalation Example)

- Impact Score: 7.7 → 9.9 (escalation)
- CET1 Ratio: 13.5% → 9.68% (at peak)
- Risk Factor Directions: 8 / 9 correct

4

## Technology Stack (Actual Implementation)



Python  
(pipeline, data processing)



ONNX Runtime  
(fine-tuned NLP model)



NumPy / Pandas / SciPy  
(data processing)



FastAPI  
(API & services)



HTML / JS Dashboard  
(frontend, visualisation)



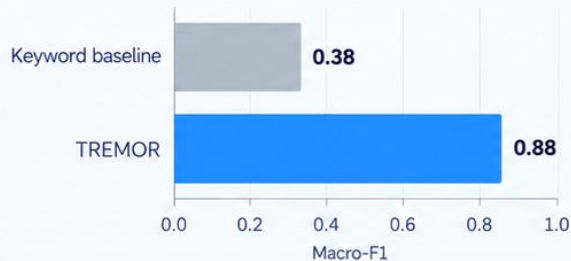
# TREMOR against the Naive Approach

Held-out data and point-in-time history only, Every number comes from docs/results/\*.json: stronger baselines and all metrics are in docs/RESULTS.md.

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## Event Type Classification

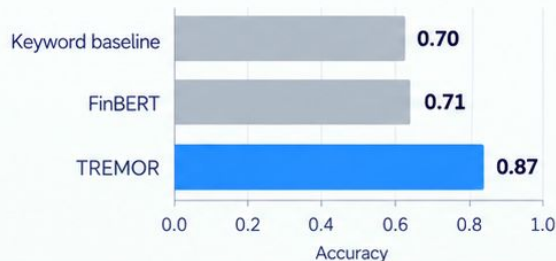
Macro-F1 on human-labelled test set



2

## Tweet Sentiment

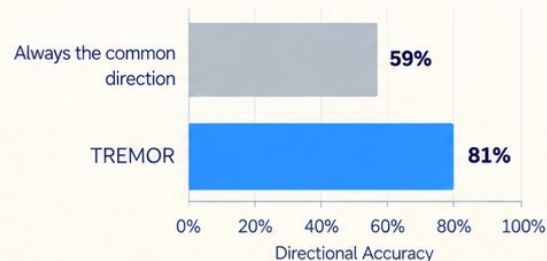
Accuracy on held-out financial tweets (TFNS)



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## Price Direction from Headlines

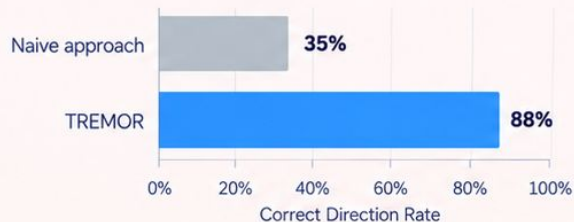
Agreement with the real move (175 headlines)



4

## Stress Scenarios: Directions

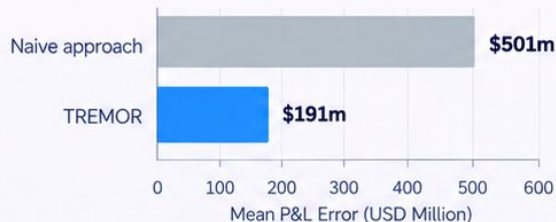
Risk-factor moves called right (21 past crises)



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## Stress Scenarios: P&L Error

Mean error in the bank's P&L (21 past crises, lower is better)



6

## Signal, Not Noise

The four-day Ukraine replay

**39,242**  
documents ingested

**14**  
stress tests out,  
each tied to a real  
development



# Case Study: Russia-Ukraine Escalation (21-24 Feb 2022)

TREMOR captures real-world events in real time, with the risk index peaking on 24 Feb 2022 and correctly identifying key risk factors.

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## Event Timeline and TREMOR Index

Key events and index movement (real time)



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## Key Risk Factor Movements

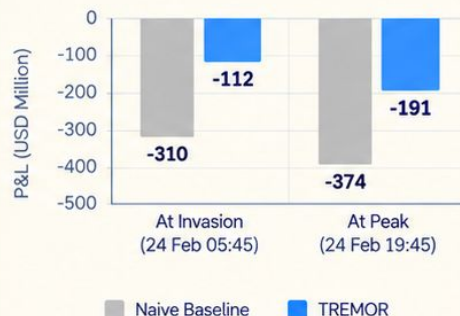
Direction predicted by TREMOR vs actual move  
(16 Feb – 8 Mar 2022)

| Risk Factor        | TREMOR Predicted Move | Actual Move | Correct? |
|--------------------|-----------------------|-------------|----------|
| Oil Prices         | ↑                     | ↑           | ✓        |
| Natural Gas Prices | ↑                     | ↑           | ✓        |
| Equity Markets     | ↓                     | ↓           | ✓        |
| Sovereign Spreads  | ↑                     | ↑           | ✓        |
| USD Index          | ↑                     | ↑           | ✓        |
| Interest Rates     | ↓                     | ↓           | ✓        |
| Credit Spreads     | ↑                     | ↑           | ✓        |
| Volatility (VIX)   | ↑                     | ↑           | ✓        |
| Gold Prices        | ↑                     | ↑           | ✓        |
| Euro (vs USD)      | ↓                     | ↓           | ✓        |

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## Portfolio P&L at Key Moments

Cumulative losses for a representative portfolio



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## Credit Early-Warning Watchlist

Flagged names before rating actions (selected examples)

| Company  | Event Date        | TREMOR Flag (Days Before) | Rating Action Date |
|----------|-------------------|---------------------------|--------------------|
| Gazprom  | 21 Feb 2022 09:15 | 4 days                    | 25 Feb 2022        |
| Sberbank | 21 Feb 2022 12:15 | 4 days                    | 25 Feb 2022        |

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## Narrative → Signals

Example event to signal extraction

Headline (24 Feb 2022, 09:20)

"Oil On The Boil: Brent races to \$105 amid Russia-Ukraine hostilities"

### Extracted Signals

Sentiment: -0.82

Event Type: Macroeconomic

Impact Score: 9.4

Price Direction: ↑ (Oil)

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## Key Takeaways

- TREMOR detects and quantifies market-moving events in real time.
- Provides accurate risk signals and factor directions.
- Enabled early identification of credit risks (e.g., Gazprom, Sberbank).
- Useful for portfolio monitoring, stress testing and scenario analysis.



# Conclusion & Future Scope

TREMOR converts unstructured news and social media into actionable, quantifiable risk signals for financial markets.

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## Key Takeaways

What we built and what we learned.



### Unstructured → Actionable Signals

Converts real-time news and social media into structured events, sentiment scores and market impact estimates.



### Improved Over Naive Baselines

Outperforms keyword and language-model baselines across event classification, sentiment analysis and market impact estimation.



### Robust and Interpretable

Provides explainable event signals, key risk factors and scenario analysis for financial decision-making.



### Practical for Risk Management

Useful for portfolio monitoring, credit watchlists and stress testing during global stress events.

2

## Use Cases & Impact

Where TREMOR can create value.



### Portfolio & Trading

Early detection of market-moving events, risk factor monitoring and scenario analysis.



### Credit Risk Monitoring

Early-warning watchlists and entity-level risk assessment for financial institutions.



### Macro & Geopolitical Risk

Track and quantify the impact of global events on markets and sectors.



### Research & Analysis

Structured and interpretable event data for economic and financial research.

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## Future Scope

Directions to further improve and extend TREMOR.



### Broader Asset Coverage

Extend to more asset classes (bonds, derivatives), more geographies and additional languages.



### Richer Event Modelling

Incorporate graph-based event propagation and multi-modal signals (audio, video).



### Production Deployment

Scale real-time streaming pipeline with continuous monitoring and reliability improvements.



### Enhanced Risk Applications

Integrate with portfolio optimization, automated trading workflows and wider credit risk use cases.

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## Project Resources

Code and documentation for reference.



**GitHub Repository** <https://github.com/ShubAce/IITKharagpur-Shubham-Suman-hackathon>

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## Author

Individual Submission



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